

SKELETAL MUSCLE

- Multi-nucleated cells
- The nuclei are located right under the cell membrane (*sarcolemma*)
- Length: from a few millimetres to one meter

SKELETAL MUSCLE

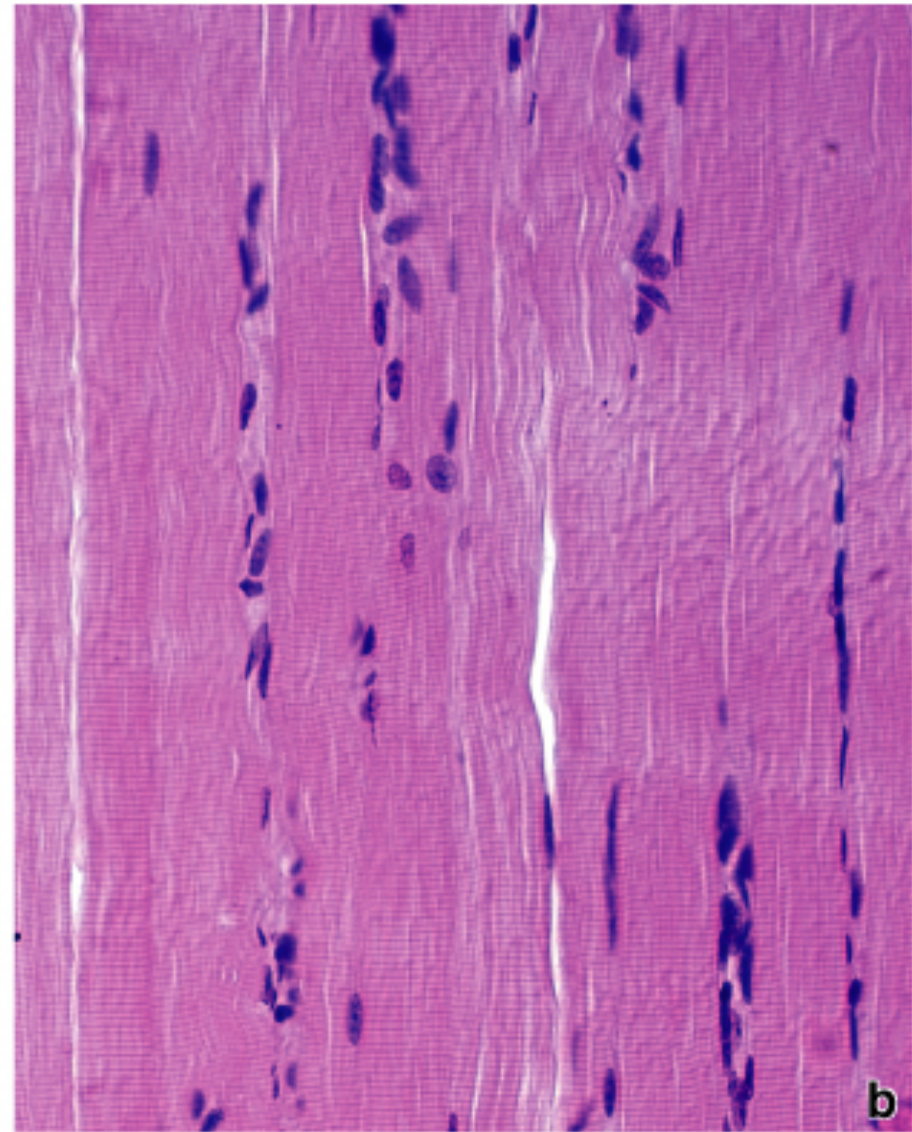
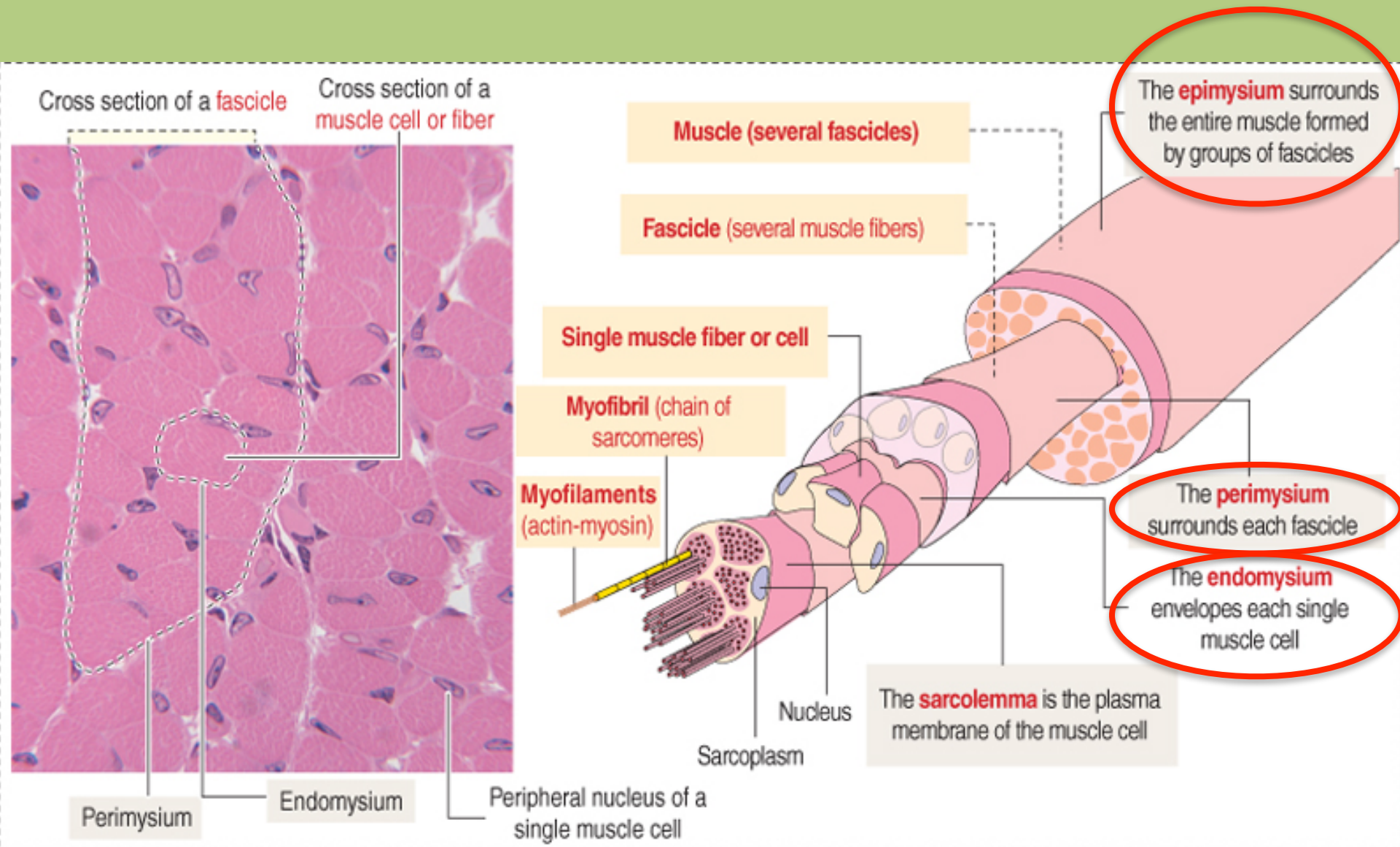


Figure 10.1b. Higher magnification photomicrograph of a skeletal muscle. X360.

Connective tissue coverings



STRIATION

A band: myosin

I band: actin

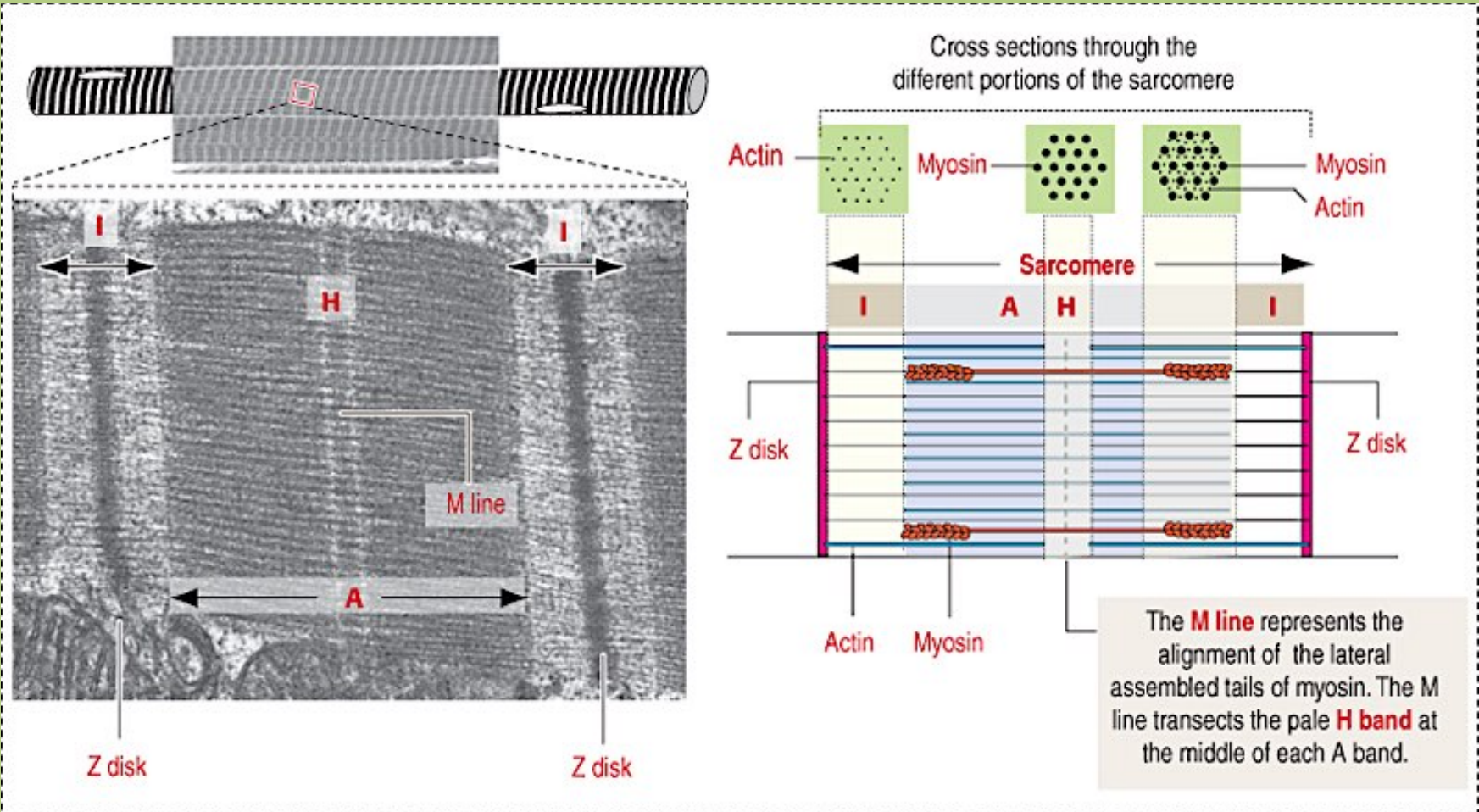
Z line: actin filaments

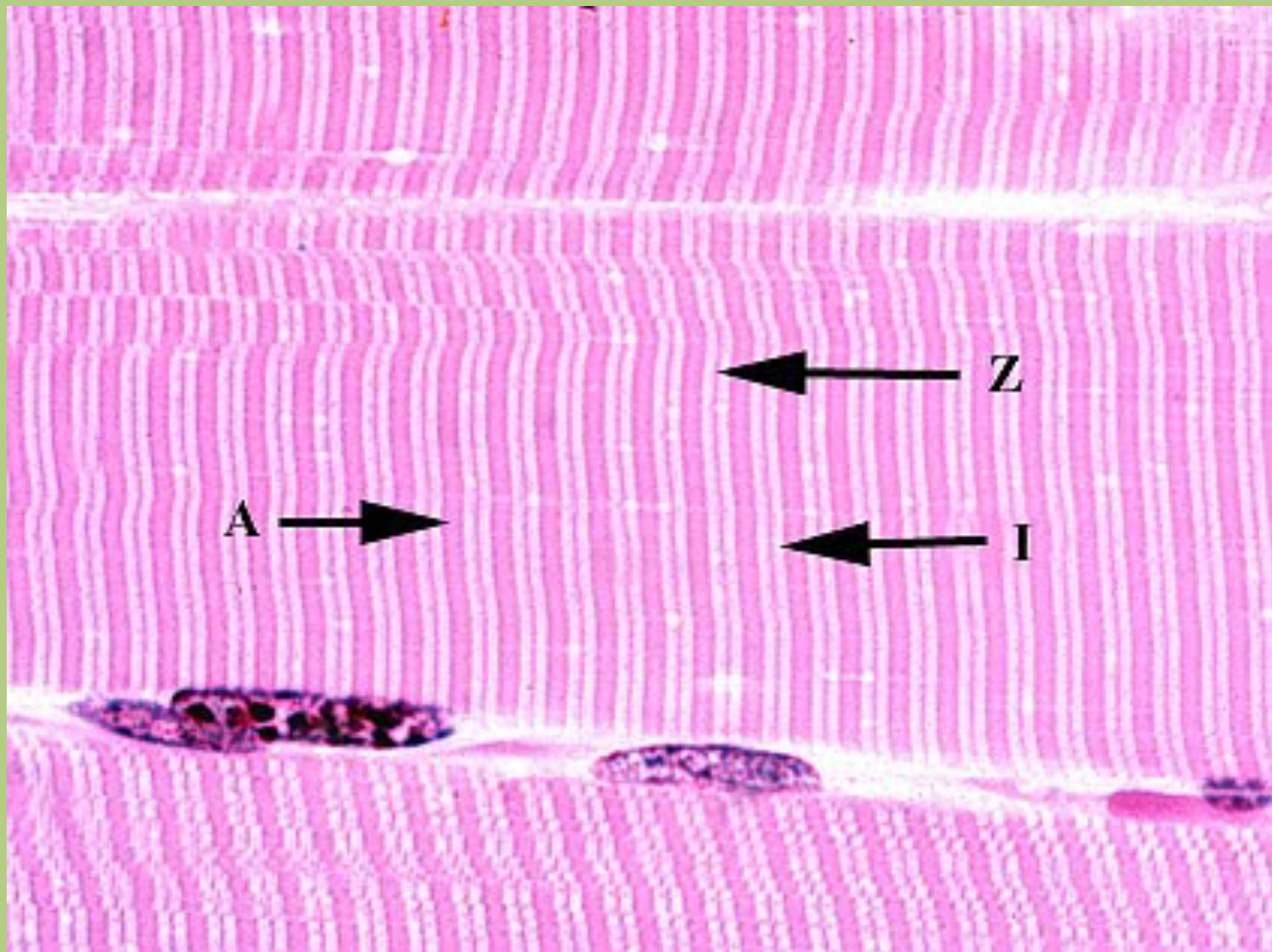
connect to it

H band: only myosin

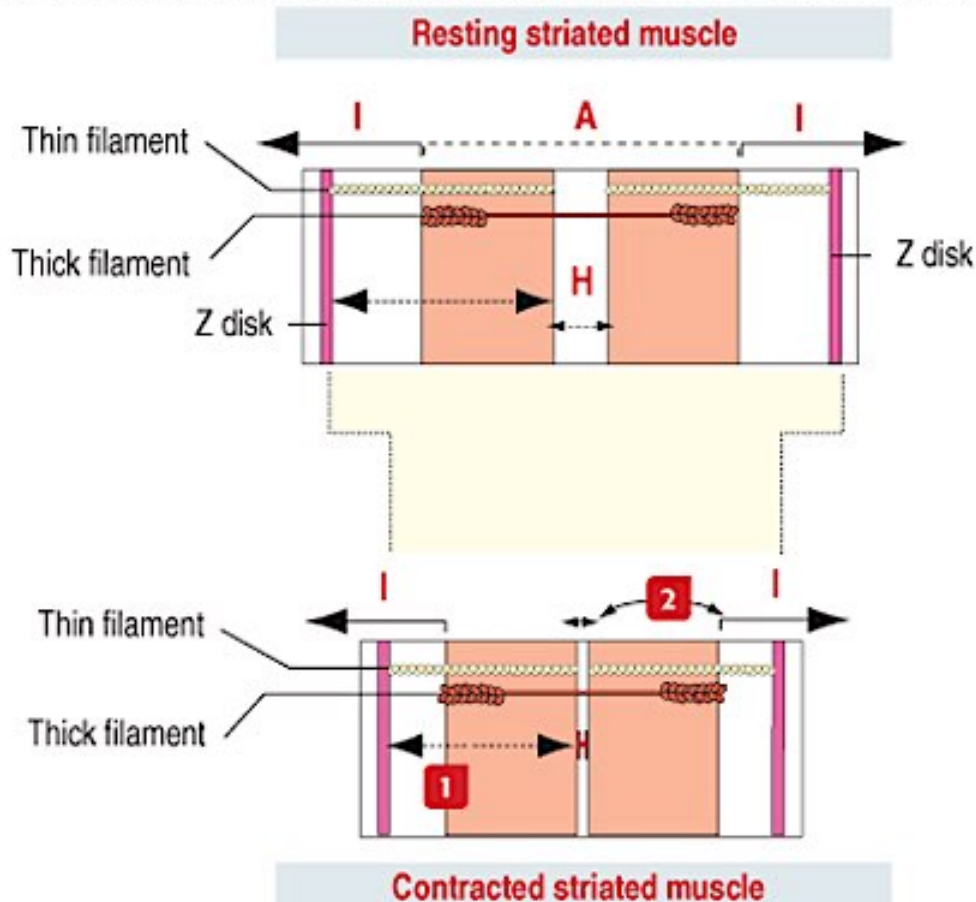
M line: myosin binding C proteins

Sarcomere: unit of the muscle tissue;
distance between two Z lines





SLIDING FILAMENT THEORY



The A band represents the distribution of the myosin thick filaments. The H band represents the myosin tail regions of the thick filaments not overlapping with thin filaments.

Thin filaments are attached to the Z disk. Two half-I bands, containing thin filaments, are seen at the right and left side of the Z disk.

During muscle contraction

1 The length of the thick and thin filaments does not change. This is demonstrated by **the constant length of the A band and the distance between the Z band and the adjacent edge of the H band.**

2 The length of the sarcomere decreases because thick and thin filaments slide past each other. This is demonstrated by **a reduction in the length of the H band and the I band.**

REGENERATION OF STRIATED MUSCLE

- Resident muscle stem cells (=satellite cells), which upon injury activate, proliferate and fuse to repair damaged fibers.

